

BASICS OF DIGITAL FORENSICS

How Hackers Hide Data in Computers

As the scope of cyber crime increases, new investigative methods using the latest scientific developments and new software are becoming popular

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The rate of cyber crime is increasing in our country. A latest survey reveals that over Rs 3500 million is lost annually due to white-collar crime, which includes cyber crime. When law enforcement agencies investigate cyber crimes, they use digital forensics, which involves investigation of computer systems to unearth evidence against cyber criminals. Many new investigative methods that use the latest scientific developments and new software are becoming popular in this field. When cyber crimes occur, they naturally involve the computer and hackers, who usually leave some evidence or tracks on the system.

Generally, cyber criminals conceal data on the hard drive. There are two popular techniques to hide data on the hard drive—one, by

low-level formatting and the other by partitioning. With low-level formatting, hackers prefer redundant sectors and bad sectors. While partitioning, data can be hidden in the inter-partition gaps, unallocated spaces, hidden partitions, boot records, partition tables and deleted partitions. To fully understand this, we should know the details of the Windows file system or other popular file systems.

Understanding file systems

Low-level formatting is performed at the factory, which creates sectors. Each sector holds 512 bytes and some overhead, which provides error correc-

tions. The hard-drive controller remaps bad sectors by redundant sectors. Criminals can misuse these redundant sectors. Fig. 1 shows details of the partitioning of the hard disk. The master boot record (MBR), which starts at sector one, consists of the master boot code (MBC) and master partition table (MPT). The volume boot record (VBR) consists of the volume boot code and disk parameter block.

File allocation table (FAT) of file systems is popular in the Microsoft Windows environment. It consists of VBR, FATs, root directory and user data area. Fig. 2 shows details of the FAT12/16 file system. FAT12 starts from the boot sector one and the directory starts from sector 20.

Fig. 3 shows the FAT32 structure where the DOS boot record makes a double copy of the boot record to maintain fault tolerance. FAT starts from the boot sector. There are various tools such as Winhex, Undelete and

File Scavenger to analyse FAT information.

NTFS is a popular file system in the Windows XP, Windows NT and Windows Server operating system. It provides compressed file features and encrypted file system features, including file system metadata about the structure of the file system.

The master file table (MFT) is the heart of NTFS. Every file

